

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

Takashi Yoshino

JCDG³ 2026

Toyo University, Japan
tyoshino@toyo.jp

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

Face-Rotation BFS Nets of the
Six Regular Convex 4-Polytopes

Takashi Yoshino
JCDG³ 2026
Toyo University, Japan
tyoshino@toyo.jp

0:20 (cumulative 0:20)

I am Takashi Yoshino, Toyo University. This talk is about nets of 4-dimensional polytopes. It closes three open cases.

Unfolding a 4-polytope

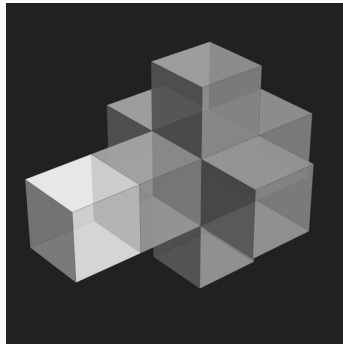
A **net** of a convex polytope: cut it open and lay the facets out flat, without overlap.

In 3D this is Dürer's problem, still open.

In 4D the facets are **3-dimensional cells** and the net lives in $\mathbb{R}^3$.

- Turney (1984): all 261 nets of the 8-cell
- Devadoss–Harvey (2023): *every* spanning tree works for the 5-, 8-, 16-cell

Left open: 24-cell, 120-cell, 600-cell.



A net of the 8-cell: Dalí's cross

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

└ Unfolding a 4-polytope

1:20 (cumulative 1:40)

Cut it open, lay it flat, no overlap. In 3D this is Dürer's problem. In 4D the pieces are solids, so the net is a 3D object you can print.

The picture: the 8-cell is the tesseract, and our net is Dalí's cross — one of Turney's 261. It is forced, not luck. Two cells are adjacent unless they are opposite, so the layers are 1 + 6 + 1 and all 6 trees give this cross.

Devadoss–Harvey settled the three small ones for *every* spanning tree. End on the last line: three polytopes left open. That is where the talk starts.

Click the picture to make it turn. Do not wait for it.

Unfolding a 4-polytope

A net of a convex polytope: cut it open and lay the facets out flat, without overlap. In 3D this is Dürer's problem, still open. In 4D the facets are **3-dimensional cells** and the net lives in $\mathbb{R}^3$.

- Turney (1984): all 261 nets of the 8-cell
- Devadoss–Harvey (2023): *every* spanning tree works for the 5-, 8-, 16-cell

Left open: 24-cell, 120-cell, 600-cell.



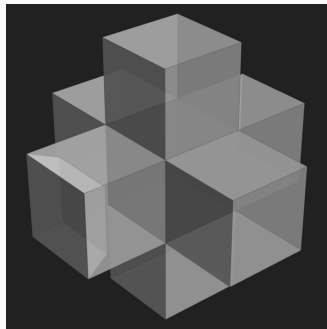
A net of the 8-cell: Dalí's cross

Face-rotation BFS

Fix a **root cell** r . Three steps.

- 1 Face-up.** Rotate in $SO(4)$ so the centroid of r sits on the $+w$ axis.
- 2 BFS.** Take a breadth-first spanning tree of the cell-adjacency graph from r . For each tree edge (parent p , child c), rotate c about the shared polygon until it lands in p 's hyperplane.
- 3 Project.** All cells now share one w -coordinate (spread $< 4 \times 10^{-13}$). Drop w .

The result is a net iff no two cells share interior volume.



Opening one BFS layer at a time, into Dalí's cross

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

└ Face-rotation BFS

1:20 (cumulative 3:00)

The slowest slide. The rotation is forced: the shared polygon is the hinge. Step 3 is a check, not an assumption — the cells really do land in one hyperplane. Do not dwell on step 2; the next slide is about what it leaves open.

Click the picture and let it run while you read the three steps. It opens *by layer*: the six neighbours first, then a pause, then the last cell. On that pause, say that BFS fixes the layers.

If asked why the cells look bent: it is a 4D view, and it fades out as the net flattens. The last frame is the real net.

Face-rotation BFS

Fix a **root cell** r . Three steps.

- 1 Face-up.** Rotate in $SO(4)$ so the centroid of r sits on the $+w$ axis.
- 2 BFS.** Take a breadth-first spanning tree of the cell-adjacency graph from r . For each tree edge (parent p , child c), rotate c about the shared polygon until it lands in p 's hyperplane.
- 3 Project.** All cells now share one w -coordinate (spread $< 4 \times 10^{-13}$). Drop w .

The result is a net iff no two cells share interior volume.



Opening one BFS layer at a time, into Dalí's cross

Step 2 hides a choice

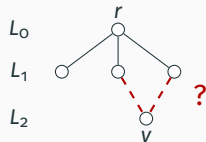
The layers are forced: L_d is the set of cells at distance d from r , and no rule can change that.

The **parent** is not. A cell in L_d may touch several cells of L_{d-1} , and BFS takes whichever one the implementation happens to reach first.

A geometric rule need not help

For the 5-, 8-, 16- and 24-cell the symmetries fixing r and v permute the candidates transitively, so **every** equivariant score — height along w , turning angle, the volume form — ties on all of them.

Ours breaks the tie by cell index, which is not equivariant either. **We come back to this.**



Two admissible parents
for v

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

Step 2 hides a choice

0:50 (cumulative 3:50)

Short, but this is the slide the referees care about. Two beats. One: BFS fixes the layers, not the parents. Two: a cleverer geometric rule does not help. For those four the candidates are interchangeable under the symmetry group, so any equivariant score ties on all of them.

Checked on every cell with more than one candidate: 1, 11 and 7 cells for the 8-, 16- and 24-cell, each a single orbit. The 5-cell has none.

If pressed: it fails for the 120-cell, where 50 of the 71 such cells have candidates in two or more orbits. Do not volunteer this.

Close with “we come back to this”.

Step 2 hides a choice

The layers are forced: L_d is the set of cells at distance d from r , and no rule can change that. The **parent** is not. A cell in L_d may touch several cells of L_{d-1} , and BFS takes whichever one the implementation happens to reach first.

A geometric rule need not help

For the 5-, 8-, 16- and 24-cell the symmetries fixing r and v permute the candidates transitively, so every equivariant score — height along w , turning angle, the volume form — ties on all of them.

Ours breaks the tie by cell index, which is not equivariant either. **We come back to this.**



Two admissible parents
for v

The validity test, and a trap

A net is valid if for every pair of cells not joined by a tree edge,

$$\text{Vol}_3(\text{conv}(\mathbf{c}_i) \cap \text{conv}(\mathbf{c}_j)) \leq 10^{-8}.$$

Boundary contact is not overlap

Testing with closed-region membership instead counts a shared face, edge or vertex as an intersection. That is common for pointed cells, and it wrongly reports

16-cell 0/16, 24-cell 0/24, 600-cell 0/600.

Measuring *volume* ignores contact of measure zero.

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

└ The validity test, and a trap

1:20 (cumulative 5:10)

Tell it as something that happened. The first run said these three polytopes failed completely.

But cells that meet along an edge do not overlap. The fix is one line of code. Without it the theorem comes out backwards.

The validity test, and a trap

A net is valid if for every pair of cells not joined by a tree edge,

$$\text{Vol}_3(\text{conv}(\mathbf{c}_i) \cap \text{conv}(\mathbf{c}_j)) \leq 10^{-8}.$$

Boundary contact is not overlap

Testing with closed-region membership instead counts a shared face, edge or vertex as an intersection. That is common for pointed cells, and it wrongly reports

16-cell 0/16, 24-cell 0/24, 600-cell 0/600.

Measuring volume ignores contact of measure zero.

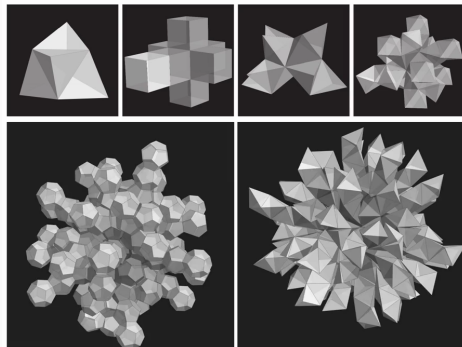
Main result

Theorem

For every regular convex 4-polytope and **every** root cell, the face-rotation BFS net is valid.

$5 + 8 + 16 + 24 + 120 + 600 = 773$ roots, all verified.

New beyond
Devadoss–Harvey: the 24-,
120- and 600-cell.



2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

└ Main result

1:40 (cumulative 6:50)

State the theorem, then let the picture sit. Six polytopes, one net each, all from root cell 1.

Say the number out loud: 773 roots, no exceptions. The overlap count is exactly zero every time — there are no near misses.

The three new ones are the three Devadoss–Harvey left open.

Click the figure and the two big nets turn, both at once. Eight seconds a turn. Let them run while you talk.

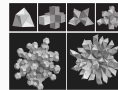
Main result

Theorem

For every regular convex 4-polytope and **every** root cell, the face-rotation BFS net is valid.

$5 + 8 + 16 + 24 + 120 + 600 = 773$ roots, all verified.

New beyond
Devadoss–Harvey: the 24-,
120- and 600-cell.



Contrast: the apple-peel algorithm

A different family of spanning trees — a greedy spiral/zonal ordering (Yoshino–Chaidee).

Polytope	BFS valid	apple-peel (RZ)	
		orderings	nets
120-cell	all roots	1440/1440	0/1440
600-cell	all roots	0/2400	—

The 120-cell has an ordering for every start and **never** a net; the 600-cell has no ordering at all.

Consequence

Having a valid *ordering* and having a valid *net* are independent properties.

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

└─ Contrast: the apple-peel algorithm

1:30 (cumulative 8:20)

The companion algorithm peels the polytope like an apple, one cell at a time.

Read the two rows as opposite failures. The 120-cell always finds an order, and the order is never a net. The 600-cell never finds an order at all.

Then the block: ordering and net are independent. Do not explain how the apple-peel rule works. Only the contrast matters.

Contrast: the apple-peel algorithm

A different family of spanning trees — a greedy spiral/zonal ordering (Yoshino–Chaidee).

Polytope	BFS valid	apple-peel (RZ)	
		orderings	nets
120-cell	all roots	1440/1440	0/1440
600-cell	all roots	0/2400	—

The 120-cell has an ordering for every start and **never** a net; the 600-cell has no ordering at all.

Consequence

Having a valid ordering and having a valid net are independent properties.

And more: the tie-breaking question

The abstract left this open:

“Whether validity extends to all BFS tie-breakings ... remains open.”

BFS layers do not depend on the rule, so a BFS tree is just a choice of parent, one layer up, for each cell — and the trees can be counted and enumerated with no rule at all.

1,132,661 trees, no self-intersection. For the first four the rule is irrelevant — including the 24-cell. And BFS is not a vacuous restriction: the 24-cell is **not** all-net (Seamons 2026) — some spanning tree does overlap.

	trees / root	checked
5-cell	1	all
8-cell	6	all
16-cell	20,736	all
24-cell	32,768	all
120-cell	$\sim 1.8 \times 10^{44}$	20 random
600-cell	$\sim 1.8 \times 10^{65}$	20 random

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

└ And more: the tie-breaking question

1:50 (cumulative 10:10)

Quote the abstract first. This answers the question the abstract asked; it is not a side remark.

The idea: the layers depend only on distance from the root, so a BFS tree is one parent per cell, taken from the layer above. The family can then be counted without naming a rule. Four polytopes exhausted, two sampled. If asked why only 20 per root: 10^{44} .

The last sentence is the payoff. It answers “why BFS and not every tree?”. For the 24-cell some spanning tree overlaps, and yet all 786,432 BFS trees do not. Say it as the reason, not as an apology.

Source, if asked: a Zenodo preprint of August 2026. I reproduced the overlap myself in exact arithmetic.

And more: the tie-breaking question

The abstract left this open:
“Whether validity extends to all BFS tie-breakings ... remains open.”

	trees / root	checked
5-cell	1	all
8-cell	6	all
16-cell	20,736	all
24-cell	32,768	all
120-cell	$\sim 1.8 \times 10^{44}$	20 random
600-cell	$\sim 1.8 \times 10^{65}$	20 random

BFS layers do not depend on the rule, so a BFS tree is just a choice of parent, one layer up, for each cell — and the trees can be counted and enumerated with no rule at all.

1,132,661 trees, no self-intersection. For the first four the rule is irrelevant — including the 24-cell. And BFS is not a vacuous restriction: the 24-cell is **not** all-net (Seamons 2026) — some spanning tree does overlap.

△ Warning: this does not extend

Beyond the regular case the rule stops being harmless.

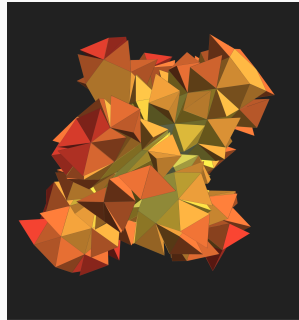
Grand antiprism, 320 cells

rule A valid from **all 320** roots
rule B valid from **no** root

Same polytope, same roots, same family of BFS trees. Only the choice of parent differs.

Among the 64 convex uniform 4-polytopes this is not isolated: another swings from 1.8% to 99.8% of its roots.

What regularity buys is not that some tree works, but that the choice does not matter.



Grand antiprism, one valid net

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

△ Warning: this does not extend

1:40 (cumulative 11:50)

This follows from the last slide, so say “and this is exactly what fails once you leave the regular case”.

The grand antiprism is the cleanest case: all 320 roots or none, decided by nothing but which parent you pick.

Then the closing sentence. Say it slowly; it is the line to remember. B2 has the full rule comparison.

△ Warning: this does not extend

Beyond the regular case the rule stops being harmless.

Grand antiprism, 320 cells

rule A valid from all 320 roots
rule B valid from no root

Same polytope, same roots, same family of BFS trees. Only the choice of parent differs.

Among the 64 convex uniform 4-polytopes this is not isolated: another swings from 1.8% to 99.8% of its roots.

What regularity buys is not that some tree works, but that the choice does not matter.



Grand antiprism, one valid net

- Why? A proof that does not enumerate.
- From BFS trees to **all** spanning trees — settled for the 24-cell (Seamons 2026), still open for the 120- and 600-cell.
- For the 120- and 600-cell even the BFS family is open: 10^{44} and 10^{65} trees per root put enumeration out of reach, so an argument would have to be structural.
- What separates the uniform polytopes that are insensitive to the rule from those that are not?

Code, data and nets:

`github.com/takashi-randomwalker/apple-peel-4d`

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

└ Open problems

0:40 (cumulative 12:30)

Do not read all four. Say the first and the last, point at the URL, and stop. The slot ends at 16:57, so this leaves about two minutes for questions.

Backups: B1 reachable trees, B2 the three rules, B3 the runcinated 24-cell, B4 what does *not* separate them, B5 the data of the six.

Open problems

- Why? A proof that does not enumerate.
- From BFS trees to **all** spanning trees — settled for the 24-cell (Seamons 2026), still open for the 120- and 600-cell.
- For the 120- and 600-cell even the BFS family is open: 10^{44} and 10^{65} trees per root put enumeration out of reach, so an argument would have to be structural.
- What separates the uniform polytopes that are insensitive to the rule from those that are not?

Code, data and nets:
`github.com/takashi-randomwalker/apple-peel-4d`

Backup: “BFS tree” is bigger than what BFS returns

A traversal reaches a tree iff its parent choices admit a total order, i.e. iff the constraint digraph $p(v) \rightarrow q$ is acyclic.

	BFS trees	reachable	
8-cell	6	6	100%
16-cell	20,736	4,896	23.6%
24-cell	32,768	2,592	7.9%

So of the 786,432 trees verified for the 24-cell, only 62,208 are reachable by any traversal. The obstruction is a *cycle* in the constraints, not merely two cells sharing two candidate parents: the 16-cell has no such pair and still leaves three quarters of its trees unreachable.

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

Backup: “BFS tree” is bigger than what BFS returns

Backup: “BFS tree” is bigger than what BFS returns

A traversal reaches a tree iff its parent choices admit a total order, i.e. iff the constraint digraph $p(v) \rightarrow q$ is acyclic.

	BFS trees	reachable	
8-cell	6	6	100%
16-cell	20,736	4,896	23.6%
24-cell	32,768	2,592	7.9%

So of the 786,432 trees verified for the 24-cell, only 62,208 are reachable by any traversal. The obstruction is a cycle in the constraints, not merely two cells sharing two candidate parents: the 16-cell has no such pair and still leaves three quarters of its trees unreachable.

Backup: which rules were compared

	queue	max-index	min-index
Runcinated 24-cell	0/240	0/240	0/240
Runcitruncated 600-cell	47/2640	2635/2640	2627/2640
Runcitruncated 120-cell	2570/2640	2640/2640	2151/2640
Grand antiprism	320/320	320/320	0/320

“queue” is the textbook BFS loop, neighbours scanned in index order.
“min/max-index” choose, for each cell, the smallest or largest indexed candidate parent.

Uniformly random trees are harsher still: of the 61 uniform polytopes valid from every root under the queue rule, 13 fail within 10 random trees, the grand antiprism among them — and every one of the four polytopes with 2,640 cells fails all ten, whether or not it is one of the 61.

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

Backup: which rules were compared

Backup: which rules were compared			
	queue	max-index	min-index
Runcinated 24-cell	0/240	0/240	0/240
Runcitruncated 600-cell	47/2640	2635/2640	2627/2640
Runcitruncated 120-cell	2570/2640	2640/2640	2151/2640
Grand antiprism	320/320	320/320	0/320

“queue” is the textbook BFS loop, neighbours scanned in index order.
“min/max-index” choose, for each cell, the smallest or largest indexed candidate parent.

Uniformly random trees are harsher still: of the 61 uniform polytopes valid from every root under the queue rule, 13 fail within 10 random trees, the grand antiprism among them — and every one of the four polytopes with 2,640 cells fails all ten, whether or not it is one of the 61.

Backup: the one that no rule could unfold

Runcinated 24-cell $t_{0,3}\{3, 4, 3\}$, 240 cells, 0/240 under all three index rules. Ask instead for a *symmetric tree*: a BFS tree invariant under $H \leq \text{Stab}(r)$ exists iff every cell has a candidate parent fixed by its stabilizer in H . Here $|\text{Aut}| = 2,304$, with two cell orbits, and the largest such H has order 3.

root orbit	$ \text{Stab}(r) $	$\max H $	H -invariant	random
48 (octahedra)	48	3	70/600	0/600
192 (triangular prisms)	12	3	83/600	0/600

The control is the point: unconstrained trees at those roots give nothing. Transitivity on each orbit then gives a valid net from **all 240 roots** — not a polytope without a net, but one whose nets no index rule finds.

(No conflict with slide 4: there the candidates form one orbit under the *full* stabilizer, so no invariant tree exists at all.)

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

└ Backup: the one that no rule could unfold

Backup: the one that no rule could unfold

Runcinated 24-cell $t_{0,3}\{3, 4, 3\}$, 240 cells, 0/240 under all three index rules. Ask instead for a *symmetric tree*: a BFS tree invariant under $H \leq \text{Stab}(r)$ exists iff every cell has a candidate parent fixed by its stabilizer in H . Here $|\text{Aut}| = 2,304$, with two cell orbits, and the largest such H has order 3.

root orbit	$ \text{Stab}(r) $	$\max H $	H -invariant	random
48 (octahedra)	48	3	70/600	0/600
192 (triangular prisms)	12	3	83/600	0/600

The control is the point: unconstrained trees at those roots give nothing. Transitivity on each orbit then gives a valid net from **all 240 roots** — not a polytope without a net, but one whose nets no index rule finds.
(No conflict with slide 4: there the candidates form one orbit under the full stabilizer, so no invariant tree exists at all.)

Backup: what does *not* separate them

Cell degree fails both ways. The grand antiprism and the snub 24-cell have cells of degree 4 — the poverty that stops apple-peel on the 600-cell — yet are valid from every root, while the runcinated 24-cell has minimum degree 5 and is valid from none. Nor does high degree help: every prism has a cell of degree up to 92 and all 17 are fine, and the root-dependent runcitruncated 120-cell carries cells of degree 32.

Angular defect at an edge — 2π less the dihedral angles meeting there, zero exactly when those cells tile $\mathbb{R}^3$ around it. For the regular six it is one number, and it orders them *exactly* by how hard they proved to be: 148.4° , 90° , 77.9° , 31.6° , 10.3° , 7.4° . Among the uniform ones it breaks both ways: the omnitruncated 120-cell shares its minimum defect, 0.736° , with the root-dependent runcitruncated 120-cell and is valid from every root, while the runcinated 24-cell sits at 10.5° , above many that are. Weighting by depth or by size does not repair it.

2026-09-06

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

└ Backup: what does *not* separate them

Backup: what does not separate them

Cell degree fails both ways. The grand antiprism and the snub 24-cell have cells of degree 4 — the poverty that stops apple-peel on the 600-cell — yet are valid from every root, while the runcinated 24-cell has minimum degree 5 and is valid from none. Nor does high degree help: every prism has a cell of degree up to 92 and all 17 are fine, and the root-dependent runcitruncated 120-cell carries cells of degree 32.

Angular defect at an edge — 2π less the dihedral angles meeting there, zero exactly when those cells tile $\mathbb{R}^3$ around it. For the regular six it is one number, and it orders them exactly by how hard they proved to be: 148.4° , 90° , 77.9° , 31.6° , 10.3° , 7.4° . Among the uniform ones it breaks both ways: the omnitruncated 120-cell shares its minimum defect, 0.736° , with the root-dependent runcitruncated 120-cell and is valid from every root, while the runcinated 24-cell sits at 10.5° , above many that are. Weighting by depth or by size does not repair it.

Backup: the six polytopes

Polytope	Symbol	V	F	C	Cell	deg	depth	trees / root
5-cell	$\{3, 3, 3\}$	5	10	5	tetrahedron	4	1	1
8-cell	$\{4, 3, 3\}$	16	24	8	cube	6	2	6
16-cell	$\{3, 3, 4\}$	8	32	16	tetrahedron	4	4	20,736
24-cell	$\{3, 4, 3\}$	24	96	24	octahedron	8	3	32,768
120-cell	$\{5, 3, 3\}$	600	720	120	dodecahedron	12	5	1.8×10^{44}
600-cell	$\{3, 3, 5\}$	120	1200	600	tetrahedron	4	15	1.8×10^{65}

deg counts the cells sharing a 2-face; *depth* is the eccentricity of the cell-adjacency graph, the same from every cell. The layers are 1+4; 1+6+1; 1+4+6+4+1; 1+8+14+1; 1+12+32+42+32+1; and sixteen layers for the 600-cell.

The 24-cell row tells the two families apart: every BFS tree of it fits in $1+8+14+1$, whereas the overlapping spanning tree of Seamons puts 13 of its 24 cells deeper than their distance from the root.

Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

- Backup: the six polytopes

Backup: the six polytopes

Polytope	Symbol	V	F	C	Celli	deg	depth	roots / root
5-cell	(3,3,3)	5	10	5	tetrahedron	4	1	1
8-cell	(4,3,3)	16	24	8	cube	6	2	6
10-cell	(3,3,4)	8	32	15	tetrahedron	4	4	20/25
120-cell	(3,3,5)	720	1440	720	octahedron	6	5	30/250
240-cell	(3,3,5)	6000	7200	3600	dodecahedron	10	5	1.8×10^{14}
800-cell	(3,3,5)	1200	1200	600	tetrahedron	4	5	1.8×10^{19}

deg counts the cells sharing a σ -face; depth is the eccentricity of the cell-adjacency graph, the same from every cell. The layers are 14c2: 16451; 14c4:64444; 148c444; 15c2:32422; 15c4:32422 and sixteen layers of the 600-cell.

The 24c-cell row tells the two families apart: every BFS tree of it fits in 148c444+1, whereas the overlapping spanning tree of Seamon's puts 13 of its 24c cells deeper than their distance from the root.